

Itumeleng Ronald Jr Modikoe

Junior Data Scientist | Machine Learning | SQL | Risk Analytics

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🐙 github.com/iModikoe 🔗 <https://rj-portfolio-mu.vercel.app/>

Profile

Junior Data Scientist and final-year Bachelor of Computing student with professional experience in credit-risk analytics, predictivemodelling, SQL data pipelines and stakeholder reporting. Builds reproducible machine-learning solutions using Python,PostgreSQL, scikit-learn, XGBoost and CatBoost, with practical experience in feature engineering, model validation, scorecards,portfolio analysis and BI dashboards. Portfolio work extends into explainable default-risk prediction, time-series feature engineering,anomaly detection and remaining-useful-life modelling

Professional Experience

Junior Data Scientist, Drive24 Pty Ltd

12/2025 – Present
Centurion

- Develop **data-driven and AI-assisted solutions** using **Python, SQL and machine learning** to automate operational analysis and support customer, credit-risk and decision-making processes.
- Build reusable **SQL extraction, transformation and analytical pipelines** integrating large relational datasets from multiple operational systems.
- Develop an **AI-assisted affordability decision workflow** that combines banking transaction data, application information, pricing data and credit-risk indicators to support and automate manual customer assessments.
- Apply **machine learning and statistical modelling** using Python, XGBoost, CatBoost and logistic regression to develop predictive classification, customer-risk and lead-prioritisation models.
- Translate ambiguous business and operational problems into **structured technical solutions, data models, decision logic and measurable outcomes**.
- Perform **feature engineering, exploratory analysis and statistical analysis** across demographic, financial, employment, behavioural and credit-related data.
- Build explainable **risk scorecards and rule-based decision frameworks** from historical customer outcomes, model findings and business requirements.
- Evaluate machine-learning models using **ROC-AUC, KS statistics, confusion matrices, feature importance and out-of-sample validation** to assess predictive and ranking performance.
- Implement **data-quality validation and investigation workflows** to identify inconsistencies across source systems and improve the reliability of downstream analytical outputs.
- Collaborate directly with **technical and non-technical stakeholders** to understand operational processes, define requirements, validate solutions and communicate recommendations and limitations.
- Analyse customer, application, contract and operational datasets to identify **behavioural patterns, anomalies and opportunities for process improvement**.
- Develop Power BI, Amazon QuickSight and Excel dashboards where required to surface operational metrics, model outputs and decision-support insights.

Technical Skills

Programming & AI Development

Python, SQL, Pandas, NumPy, scikit-learn, Jupyter Notebook, R, LLM-Assisted Workflows, AI-Driven Automation

Machine Learning and Statistics

XGBoost, CatBoost, Logistic Regression, Random Forest, Decision Trees, Binary Classification, Predictive Modelling, Feature Engineering, Model Evaluation, ROC-AUC, KS Statistic, Feature Importance, Statistical Analysis, Hypothesis Testing

Data Engineering & Databases

PostgreSQL, SQL Server, T-SQL, Oracle, Relational Databases, ETL, Data Extraction, Data Transformation, Data Cleaning, Data Validation, Data Integration, Data Preparation, Views, Stored Procedures, Triggers, ERD Design

Business Intelligence & Analytics

Power BI, Amazon QuickSight, Excel, Matplotlib, Dashboard Development, KPI Reporting, Exploratory Data Analysis, Data Storytelling, Stakeholder Reporting

Development Tools & Cloud

Git, GitHub, Visual Studio Code, PyCharm, Anaconda, Talend, AWS Fundamentals

Development Tools and Platforms

Git, GitHub, Visual Studio Code, PyCharm, Anaconda, Talend, AWS Fundamentals

Projects

Explainable Credit Default Risk Prediction [🔗](#), Python, scikit-learn, Streamlit, GitHub Actions

Built an end-to-end default-risk system using 30,000 customer records. Engineered repayment and affordability indicators, compared Logistic Regression, Random Forest and HistGradientBoosting using validation PR-AUC, excluded sex from predictive inputs, evaluated subgroup performance and deployed an adjustable-threshold Streamlit application. Achieved 82.0% test accuracy, 67.3% precision and 0.779 ROC-AUC

Fleet Predictive Maintenance and Remaining Useful Life [🔗](#),

Python, Time Series, Gradient Boosting, Isolation Forest, Streamlit

Developed a NASA C-MAPSS prognostics pipeline using engine-level validation to prevent leakage. Created rolling sensor statistics and degradation-trend features, compared regression models, estimated remaining useful life and added anomaly monitoring for maintenance prioritisation. The compact deployment model achieved MAE 14.74 cycles and R^2 0.767 across 100 official test engines.

Credit Risk & Lead Prioritisation Model – Drive24,

SQL | Python | Power BI | XGBoost | CatBoost | Statistical Analysis

Developed an end-to-end customer-risk scoring and lead-prioritisation framework using historical contract outcomes, demographic information, financial variables and credit-bureau characteristics. Defined measurable Good and Bad customer outcomes, performed data cleaning and feature engineering, analysed predictive patterns using WOE/IV and trained XGBoost and CatBoost classification models.

Evaluated model performance using ROC-AUC, KS statistics, confusion matrices and ranking analysis.

Converted model findings into explainable scoring rules and stakeholder-facing reports designed to prioritise higher-potential leads for further assessment.

AI-Driven Affordability & Decision Automation – Drive24,

Python | SQL | AI/LLM Workflows | Transaction Data | Decision Automation

Developing an AI-assisted decision-support workflow to automate parts of the customer affordability assessment process using banking transaction, application, pricing and credit-risk data. Built SQL-based data pipelines to integrate multiple operational data sources and derive income stability, financial commitments, cash-flow retention and affordability indicators. Incorporated AI-assisted transaction classification and structured decision logic to transform unstructured financial activity into decision-useful features. Worked directly with operational stakeholders to translate manual assessment processes into explainable, repeatable automation aimed at reducing review effort and improving decision consistency.

IoT-Powered Mobile Weather Observation Station [🔗](#), ESP32, Sensors, IoT, Cloud Analytics, Data Visualisation

Designed and developed an IoT-based mobile weather observation station using ESP32 sensors to collect real-time environmental data. Integrated sensor outputs with cloud-based analytics and visualisation components to support weather monitoring and solar-vehicle strategy optimisation. Completed as a final-year multidisciplinary computing project.

Education

Bachelor of Computing: Business Intelligence, Belgium Campus iTversity

Current Academic Average: 78.5%

01/2023 – 12/2026

Pretoria, South Africa

Relevant Areas of Study: Data Science, Machine Learning, Statistics, Business Intelligence, Database Development, Data Warehousing, Data Visualisation, Software Development and Research Methods

Certifications

- IBM Data Science Professional Certificate [🔗](#)
- IBM Data Analyst Professional Certificate [🔗](#)
- The Data Science Course: The Complete Data Science Bootcamp 2025 [🔗](#)
- AWS Cloud Practitioner Essentials [🔗](#)
- Mastering SQL Server 2016 Integration Services (SSIS) - Part 1 [🔗](#)
- Excel 101 and Excel 202

Leadership and Achievements

Awarded a bursary to pursue a Bachelor of Computing in Business Intelligence at Belgium Campus iTversity, NTT DATA

Selected for the Saturday School Programme for outstanding academic potential in technology, Dimension Data Saturday School Programme

Peer Tutor, Belgium Campus iTversity

Supporting students with academic and technical subjects.

Former Student Representative Council member, Residence Representative and Soccer Team Captain, Belgium Campus iTversity